Sparse-Observation Reconstruction with a Modular Physics-Informed Neural Network Framework

A Reproducible Study of Heat, Waves, Beam Bending and Orbital Dynamics

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Abstract

This study tests whether a differential-equation penalty improves reconstruction from sparse observations. We evaluate the shared training engine of PINN Multi-Physics Explorer on four synthetic systems with analytical reference solutions: heat diffusion, standing waves, uniformly loaded Euler–Bernoulli beams and two-body orbital dynamics. Each physics-informed neural network (PINN) is paired with a data-only multilayer perceptron (MLP) of identical capacity and initialization, using the same observations at 5%, 15% and 30% of a candidate grid. Three seeds per setting give 72 trained models. Predictions are evaluated on a disjoint test grid, with coordinate and output scales fixed without using test labels. Evaluation includes per-output coefficient of determination, root mean squared error, relative L_2 error, separately sampled residual diagnostics and measured training time. At 15% observations, mean test relative L_2 errors are 0.0067 versus 0.0172 for heat, 0.0351 versus 0.0286 for waves, 0.0051 versus 0.0080 for beams and 0.0268 versus 0.0181 for orbital position (PINN versus MLP). The direction of the difference changes with both the equation and the observation fraction. These results evaluate an existing implementation on noise-free interpolation tasks. Their scope is limited by the fixed loss weight and the absence of explicit initial and boundary constraints.

Keywords: physics-informed neural networks; sparse observations; automatic differentiation; scientific machine learning; reproducibility.

1 Introduction

Sparse measurements do not generally determine a physical field uniquely. A differential equation provides additional structure by relating derivatives of the field at locations where no measurements exist. A PINN adds the squared equation residual to the observation loss. This penalty can narrow the set of plausible reconstructions, although finite optimization with a finite penalty cannot guarantee that the equation is satisfied everywhere.

PINN Multi-Physics Explorer is a Python application that separates the neural approximation from the equation-specific residual. It includes heat, wave, orbital, elasticity, river-flow and reservoir workflows, with a Dash interface and standalone orbital visualization pages. This paper studies four analytical benchmark systems. Together they exercise first-order time derivatives, second-order time and space derivatives, fourth-order space derivatives and coupled nonlinear orbital accelerations. The river-flow, inverse-roughness and reservoir modules are part of the software, but their predictive performance is outside this benchmark.

The study asks whether a differential-equation penalty improves off-grid reconstruction compared with an otherwise matched data-only network, and whether any improvement is consistent across

observation fractions, seeds and equation types. Using one training engine allows the same comparison across all four systems.

The contribution is an empirical evaluation of the existing implementation under a fixed, documented protocol. Paired multi-seed runs, saved predictions, model states and run configurations make the numerical figures and tables reproducible. Scores from earlier project examples are excluded from the study.

2 Related work

Raissi, Perdikaris and Karniadakis formalized a widely used continuous-time PINN approach for forward and inverse differential-equation problems [1]. Their formulation combines observational supervision with residuals obtained through automatic differentiation. Our implementation follows the same principle. The experiment turns the physics penalty on or off while holding the neural approximation and the data fixed.

Reusable differential-equation libraries already support many classes of equation. DeepXDE, for example, offers compact problem definitions and automatic differentiation for several forward and inverse settings [2]. Here, the module structure provides a common implementation for paired comparisons across four systems; the study introduces no new PINN formulation.

PINNs can also be hard to train. Wang, Teng and Perdikaris analyzed stiffness and imbalance between backpropagated loss gradients [3]. Krishnapriyan et al. demonstrated failures under several differential operators and showed that equation-informed penalties can make optimization harder [4]. These studies support reporting both observation error and residual error. The experiments here do not measure gradient conflicts or test any causal explanation for a failure.

The wider project also contains Monte Carlo dropout tools, which relate to approximate Bayesian uncertainty methods [5]. Those tools are not part of this deterministic benchmark. Variation across the three training seeds is reported as run-to-run variability. It is not a predictive uncertainty interval or a confidence guarantee.

3 Mathematical formulation

Let z denote physical model coordinates and $u(z)$ the scalar or vector state. A domain supplies an operator $\mathcal{F}$ such that

$$\mathcal{F}[u](z) = 0, \qquad r_\theta(z) = \mathcal{F}[u_\theta](z). \tag{1}$$

For N_d observations (z_i, y_i) and N_f collocation coordinates c_j, the losses are

$$L_d = \frac{1}{N_d d_u} \sum_{i=1}^{N_d} \sum_{k=1}^{d_u} (\widehat{u}_{\theta,k}(\widehat{z}_i) - \widehat{y}_{i,k})^2, \tag{2}$$

$$L_f = \sum_{k=1}^{d_r} \frac{1}{N_f} \sum_{j=1}^{N_f} r_{\theta,k}(c_j)^2, \tag{3}$$

$$L_{\mathrm{PINN}} = L_d + 0.1 L_f, \qquad L_{\mathrm{MLP}} = L_d. \tag{4}$$

Hats mark the coordinates and outputs that the network sees. The residual is evaluated through a physical-coordinate adapter, so output normalization affects L_d, while L_f keeps the equation written in the model coordinates below. Residual scales are set to one in all four benchmarks. This is a fixed choice for the experiment; residual sizes should not be compared across equations.

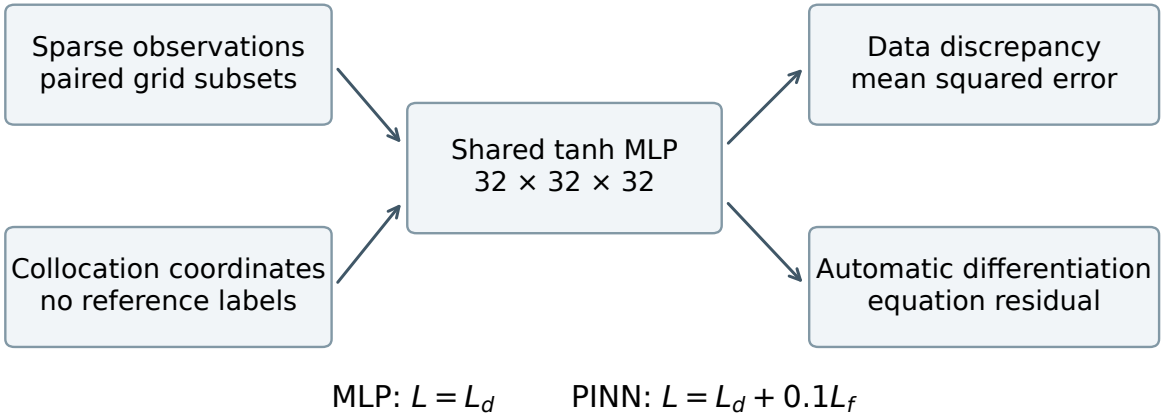


Figure 1: Shared computational structure. Both models receive the same sparse labels and initialization. Only the PINN uses residuals at unlabeled collocation coordinates. The diagram shows the configuration that was measured, not every optional architecture in the application.

3.1 Network and physical-coordinate adapter

Both models use the project's `PINNEngine`: three fully connected hidden layers of width 32, tanh activations, Xavier-normal weight initialization and zero biases. Within a domain they have identical parameter counts. The tanh activation is smooth, so it can be differentiated as many times as the residuals need. A piecewise linear activation would be unsuitable for these higher-order residuals.

If $\widehat{z} = (z - m_z)/s_z$ and $u = m_u + s_u\widehat{u}$, a derivative with respect to z must include its scale factor. In particular,

$$\frac{\partial u}{\partial z_j} = \frac{s_u}{s_{z,j}}\frac{\partial \widehat{u}}{\partial \widehat{z}_j}, \qquad \frac{\partial^4 u}{\partial x^4} = \frac{s_u}{s_x^4}\frac{\partial^4 \widehat{u}}{\partial \widehat{x}^4}. \tag{5}$$

The project's `NormalizedPDE` rebuilds physical coordinates and states inside the differentiable graph before calling each residual, which keeps the chain rule intact for higher derivatives and nonlinear terms. Each input interval is mapped to $[-1, 1]$. Output centers are zero. Output scales are one for heat, waves and both orbital coordinates, and 0.01 for beam deflection. These constants are fixed before training; none is estimated from the test reference.

3.2 Heat diffusion

The first system is

$$T_t - \alpha T_{xx} = 0, \quad (x,t) \in [0,1]^2, \quad \alpha = 0.1, \tag{6}$$

with reference solution

$$T(x,t) = \sin(\pi x)\exp(-\alpha\pi^2 t). \tag{7}$$

This reference has zero endpoint temperature and initial profile $T(x,0) = \sin(\pi x)$. The residual uses one time derivative and two space derivatives. The reference is generated by the project's analytical heat helper with unit amplitude.

3.3 Standing waves

For unit wave speed, the second system is

$$u_{tt} - u_{xx} = 0, \quad (x,t) \in [0,1]^2, \quad u(x,t) = \sin(\pi x)\cos(\pi t). \tag{8}$$

The reference describes a string fixed at both ends, with sinusoidal initial displacement and zero initial velocity. It covers half of the fundamental mode's period. Its residual needs second derivatives in both input coordinates.

3.4 Uniformly loaded beam

The beam satisfies the static fourth-order ordinary differential equation

$$EI\, v^{(4)}(x) = q_0, \quad x \in [0,1], \quad EI = q_0 = 1, \tag{9}$$

with simple-support reference conditions $v(0) = v(1) = v''(0) = v''(1) = 0$. Its exact deflection is

$$v(x) = \frac{x^4 - 2x^3 + x}{24}, \qquad v(1/2) = \frac{5}{384}. \tag{10}$$

The benchmark uses the uniform-load implementation only. The project also has a smooth Gaussian approximation of a concentrated load, but that is not used in this comparison.

3.5 Two-body orbital dynamics

The orbital state $(x(t), y(t))$ follows a coupled ordinary differential equation,

$$x'' + \frac{\mu x}{r^3} = 0, \qquad y'' + \frac{\mu y}{r^3} = 0, \qquad r^2 = x^2 + y^2, \tag{11}$$

with $\mu = 1$, semi-major axis $a = 1$, eccentricity $e = 0.3$ and $t \in [0, 2\pi]$. The reference uses the eccentric anomaly E, obtained from Kepler's equation:

$$E - e\sin E = t, \quad x = \cos E - e, \quad y = \sqrt{1-e^2}\sin E. \tag{12}$$

The project residual keeps a numerical guard, replacing r by $\sqrt{x^2 + y^2 + 10^{-6}}$. An exact, unsoftened Kepler orbit therefore has a small nonzero residual in the implementation; Section 4.4 gives its size. The orbit is a trajectory in a plane governed by an ODE, not a continuous gravitational field in space.

3.6 What selects a solution

The stated boundary and initial conditions define the analytical references, but the training protocol does not enforce them. There are no separate penalties for boundary values, initial displacement, initial velocity, support moments or orbital velocity. Both models receive the same sampled field or position labels, and the PINN also receives the residual penalty in Eq. (4). The random observation subset may include some boundary positions, but boundary coverage is not guaranteed. The task is reconstruction from sparse observations, rather than an initial–boundary-value problem with complete conditions. A small beam residual cannot pin down the cubic homogeneous component, and orbital accelerations alone do not identify an orbit.

4 Experimental design

4.1 Observation and test coordinates

Heat and wave observation candidates lie on a 25×25 endpoint-inclusive grid. Their independent test grids use 48×48 cell midpoints. The beam uses 100 endpoint-inclusive candidate positions and 500 test midpoints. The orbit uses 400 endpoint-inclusive candidate times over one period

Table 1: Fixed benchmark settings. Fractions refer to the observation candidate grid. All references are noiseless synthetic solutions in idealized model units.

System	Candidate / test points	Inputs / outputs	Parameters
Heat	625 / 2,304	(x, t) / T	$\alpha = 0.1$
Wave	625 / 2,304	(x, t) / u	$c = 1$, mode 1
Beam	100 / 500	x / v	$EI = q_0 = L = 1$
Orbit	400 / 800	t / (x, y)	$\mu = a = 1$, $e = 0.3$

and 800 test midpoints. Before conversion to single precision, the test coordinates are checked to be disjoint from the candidate coordinates. No test labels enter the loss, normalization or checkpoint selection.

For each seed, one random permutation of the candidate indices defines nested subsets at nominal fractions of 5%, 15% and 30%, using $\lfloor fN \rfloor$ observations. The counts are 31/93/187 for heat and wave, 5/15/30 for the beam and 20/60/120 for the orbit. The fraction describes locations spread throughout the domain, so 15% orbital observations are scattered over the whole orbit, not taken from its first 15%. The protocol tests interpolation between randomly placed observations. It does not measure forecasting in time or extrapolation outside the domain.

4.2 Paired model training

The three seeds are 7, 19 and 41. For a given domain, fraction and seed, the MLP and the PINN start from exactly the same initial state dictionaries and see exactly the same observations. Each model takes 2,000 full-batch Adam steps with learning rate 10^{-3}, cosine annealing to 10^{-5} and gradient-norm clipping at 1.0. The PINN draws 128 new uniform collocation coordinates per step using the project's epoch-indexed sampler. For a given domain, this collocation sequence is shared across fractions and training seeds. The PINN's residual loss sums the mean squared residual components, and its observation loss averages over observations and output components.

The plain network uses the same optimizer, scheduler and clipping but skips residual computation. The PINN calls the project's `train_pinn` function unchanged. There are no restarts, early stopping, test-selected checkpoints or per-domain hyperparameter searches. A separate 200-step pilot run checked that the code executed and wrote its output files; its results are not part of the benchmark. Fixing the physics weight and architecture makes the comparison easy to audit, but leaves each model's best achievable performance unresolved.

4.3 Evaluation metrics

For each output coordinate, with test targets y_i and predictions p_i, we report

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i (p_i - y_i)^2}, \tag{13}$$

$$R^2 = 1 - \frac{\sum_i (p_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{14}$$

$$\varepsilon_{L_2} = \frac{\sqrt{\sum_i (p_i - y_i)^2}}{\sqrt{\sum_i y_i^2}}. \tag{15}$$

The sparsity figure combines all output coordinates in the numerator and denominator of the relative L_2 error. The results table reports the two orbital coordinates separately. RMSE is in the original reference model units and is included in the accompanying CSV. Negative R^2 values are kept where they occur.

We report means across three seeds and sample standard deviations with the $n-1$ denominator. The standard deviations reflect variation from both initialization and the random choice of observations; the experiment does not separate the two. They are not confidence intervals and do not quantify measurement uncertainty. On log plots, an error bar whose lower end would cross zero is clipped to 5% of the mean for display; the exported standard deviation is unchanged.

Residual diagnostics use 512 coordinates sampled uniformly with a separate fixed random seed, 2026. The sampler does not exclude training collocation coordinates, so the diagnostic sample is not strictly disjoint from them. Residual RMS measures how well the equation is satisfied at these locations; it does not measure agreement with the reference solution. Residual sizes are not pooled across domains, and a small residual alone does not establish reconstruction accuracy.

4.4 Execution and provenance

All training runs use float32 PyTorch on an AMD Ryzen 5 PRO 2500U CPU, with two worker processes and one PyTorch thread per worker. The recorded environment is Python 3.14.8, PyTorch 2.15.0+cu132 and NumPy 2.5.4. CUDA is unavailable and no GPU is used. Per-model wall times cover optimization only. They come from concurrent CPU runs, so they are not an isolated hardware benchmark.

The source snapshot is based on project commit `f3b9ad1`, and file-level SHA-256 hashes record the working-tree files actually used. Before training, float64 residual checks on the analytical functions give maximum absolute values of 2.22×10^{-16} for heat, 1.78×10^{-15} for the wave, zero for the uniform beam and 4.33×10^{-6} for the guarded orbital residual. These checks test the residual code and scaling path, not the networks' accuracy. Twenty-eight existing checks covering normalization, the engine and domain operations also pass on the copied implementation.

5 Results

5.1 Overall comparison

Table 2 gives per-output scores at 15% observations, and Fig. 2 shows all fractions. At 15%, the PINN has lower mean relative error for heat and the beam, while the MLP has lower mean error for the wave and the orbit. The appendix gives all fraction-level summaries, and the CSV contains every per-seed and per-output measurement.

The PINN has lower relative error in 9 of 9 paired heat runs. At 5% observations, the PINN's mean error is lower in every system, but it improves on the matched MLP in only 1 of 3 wave runs and 2 of 3 orbital runs. A lower three-seed mean therefore does not imply an improvement in every paired run.

5.2 Heat and waves

For heat at 15%, the MLP's mean relative error is 2.57 times the PINN's. The PINN's mean R^2 is 0.99977, against 0.99878 for the data-only model. For the standing wave, the PINN's mean relative error is 1.23 times the MLP's at the same fraction. The wave's high PINN R^2 therefore coexists with a larger prediction error than the data-only baseline. Figure 3 shows both models and the reference for the prespecified seed 7.

5.3 Beam reconstruction

With five beam observations, MLP test R^2 ranges from -0.3476 to 0.9169 across the three seeds, while PINN R^2 ranges from 0.9994 to 0.9996. At 15 observations, mean test R^2 is 0.99980 for

Table 2: Independent test performance at the nominal 15% observation fraction. Entries are means ± sample standard deviations over three paired seeds. Higher R^2 and lower relative L_2 are better. All values were measured with the configuration in Table 1.

Domain / output	Model	Test R^2	Relative L_2
Heat diffusion	MLP	0.9988 ± 0.0005	0.0172 ± 0.0034
Heat diffusion	PINN	0.9998 ± 0.0003	0.0067 ± 0.0043
Wave propagation	MLP	0.9992 ± 0.0004	0.0286 ± 0.0059
Wave propagation	PINN	0.9987 ± 0.0006	0.0351 ± 0.0079
Beam bending	MLP	0.9996 ± 0.0003	0.0080 ± 0.0038
Beam bending	PINN	0.9998 ± 0.0002	0.0051 ± 0.0043
Orbital dynamics (x)	MLP	0.9996 ± 0.0001	0.0173 ± 0.0029
Orbital dynamics (y)	MLP	0.9996 ± 0.0001	0.0192 ± 0.0029
Orbital dynamics (x)	PINN	0.9986 ± 0.0010	0.0302 ± 0.0108
Orbital dynamics (y)	PINN	0.9995 ± 0.0003	0.0208 ± 0.0077

the PINN and 0.99960 for the MLP. The comparison uses a uniform distributed load and a fixed output scale; the concentrated-load approximation remains untested. Figure 4 shows the seed-7 deflection curves and observed points.

5.4 Orbital reconstruction

With 60 observed orbital times, the PINN's mean test R^2 is 0.9986 for x and 0.9995 for y; the matched MLP obtains 0.9996 and 0.9996, respectively. Joint relative position errors are 0.0268 and 0.0181. Reporting the coordinates separately exposes differences that an orbit-plane plot can hide. Because observations span the full period, these are reconstruction scores rather than forecasts from an initial orbital arc.

5.5 Optimization and residual diagnostics

Figure 5 plots observation fitting separately from the unweighted physics penalty, and Figure 6 evaluates residuals at separately sampled diagnostic coordinates. Residual RMS measures agreement with the differential equation at these locations; test error measures agreement with the reference solution. The appendix gives measured timing and parameter counts. The models take the same number of optimization steps, but residual computation makes each PINN step more expensive.

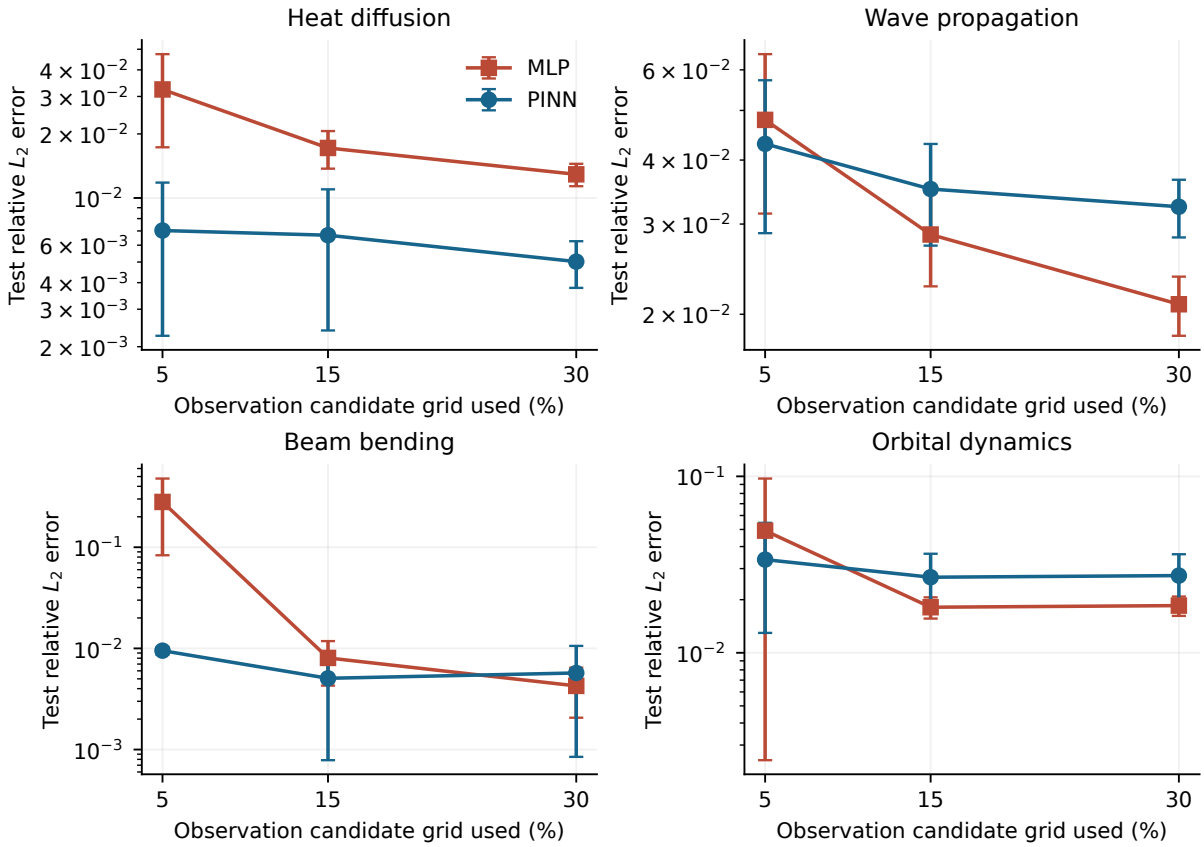


Figure 2: Off-grid reconstruction error versus observation fraction. Symbols show three-seed means and bars show sample standard deviations. All runs are included. The orbital metric combines the squared errors of both position coordinates. Training and test grids share no coordinates.

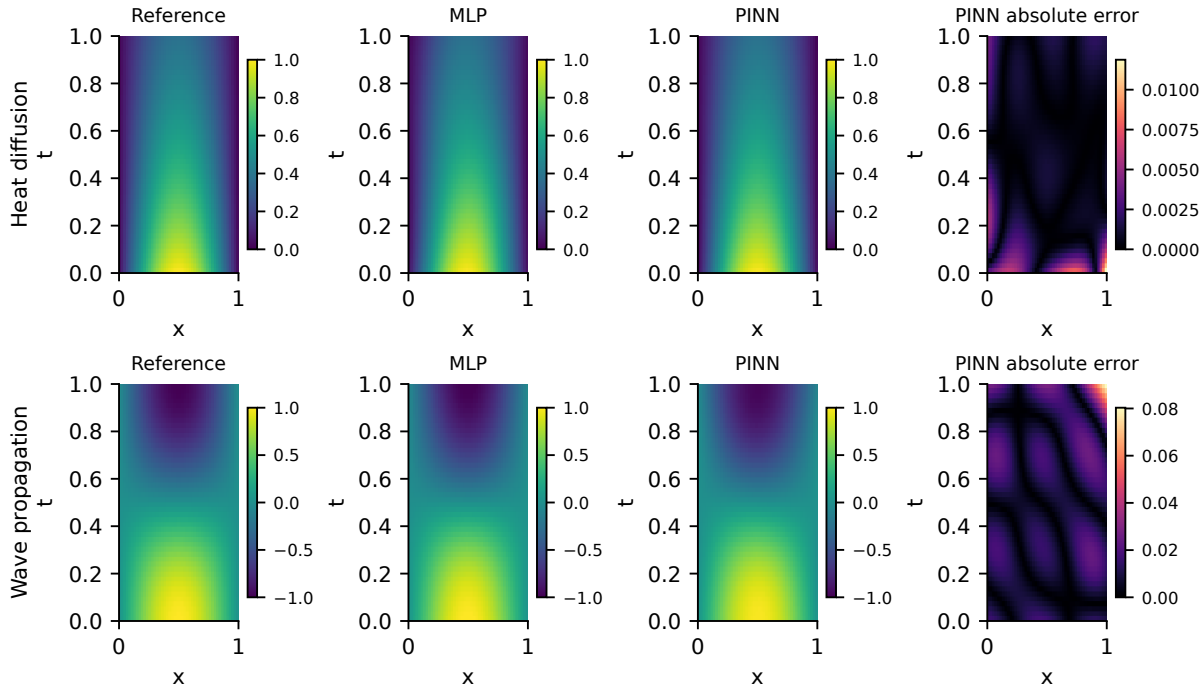


Figure 3: Heat and wave reference fields, MLP predictions, PINN predictions and PINN absolute errors for the prespecified seed 7 at 15% observations. Reference and prediction panels share a color scale within each row; the error panel has its own scale. Each panel contains the 2,304 test-grid values.

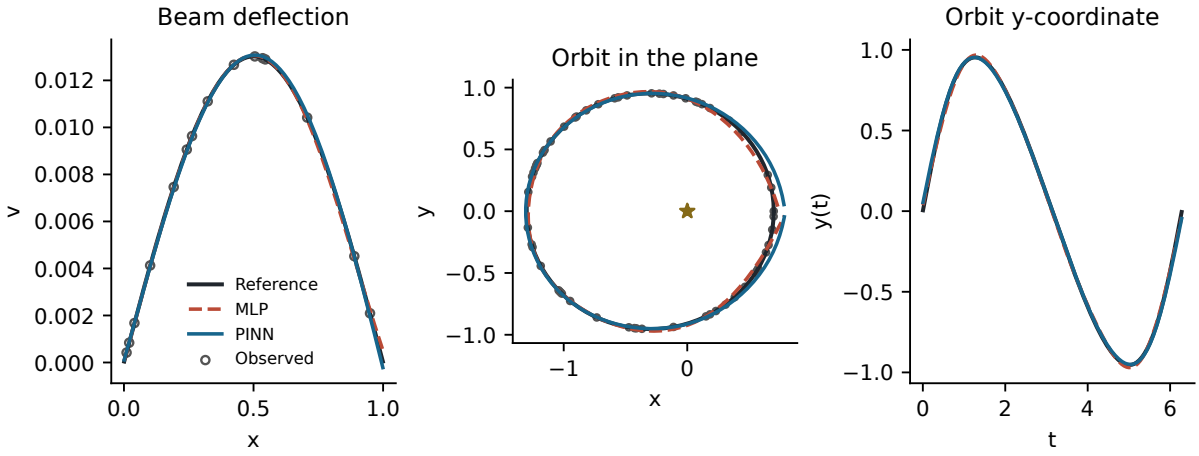


Figure 4: Uniform-beam and orbital reconstructions for seed 7 at 15% observations. Open markers are observed labels; continuous curves sample the independent test grid. The orbital observations are spread over the full period. The right panel shows the y coordinate against time, which an orbit-plane plot alone cannot show.

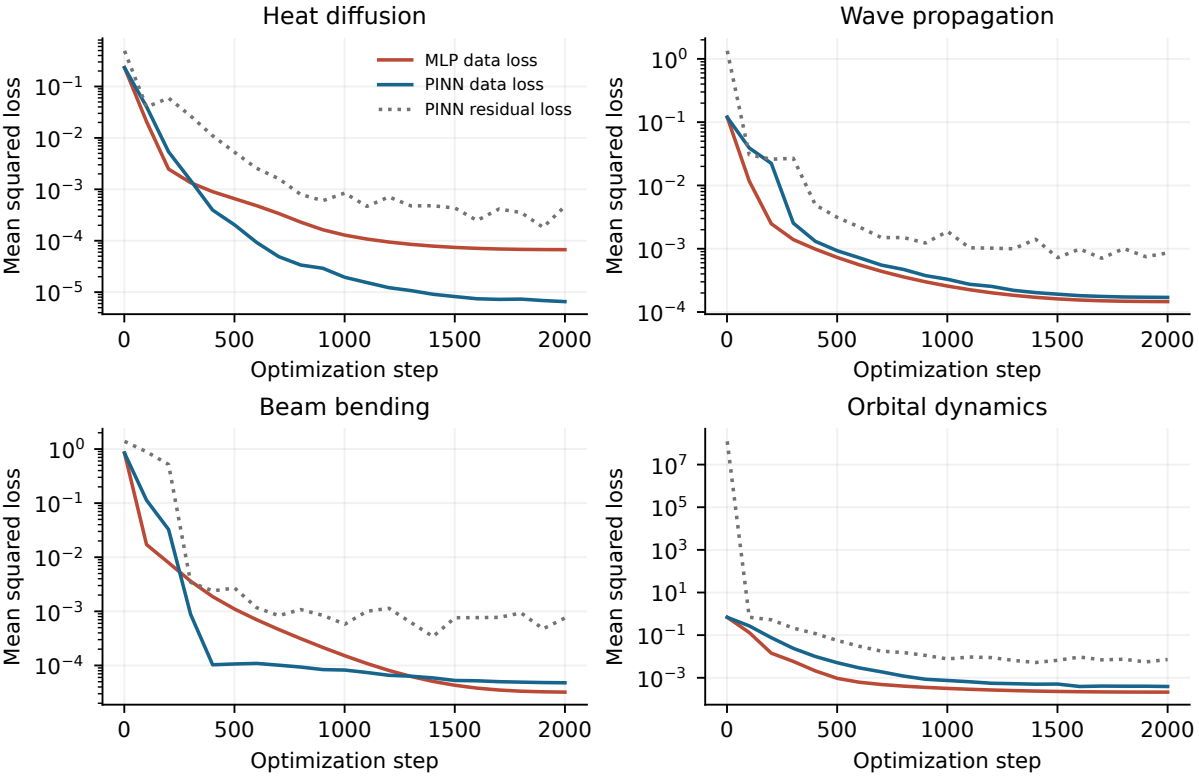


Figure 5: Training histories for seed 7 at 15% observations. Observation MSE is plotted separately from the PINN's unweighted residual loss. Equal step budgets do not mean equal wall time, because the PINN differentiates the state with respect to the coordinates. The histories are diagnostics and were not used to choose stopping points.

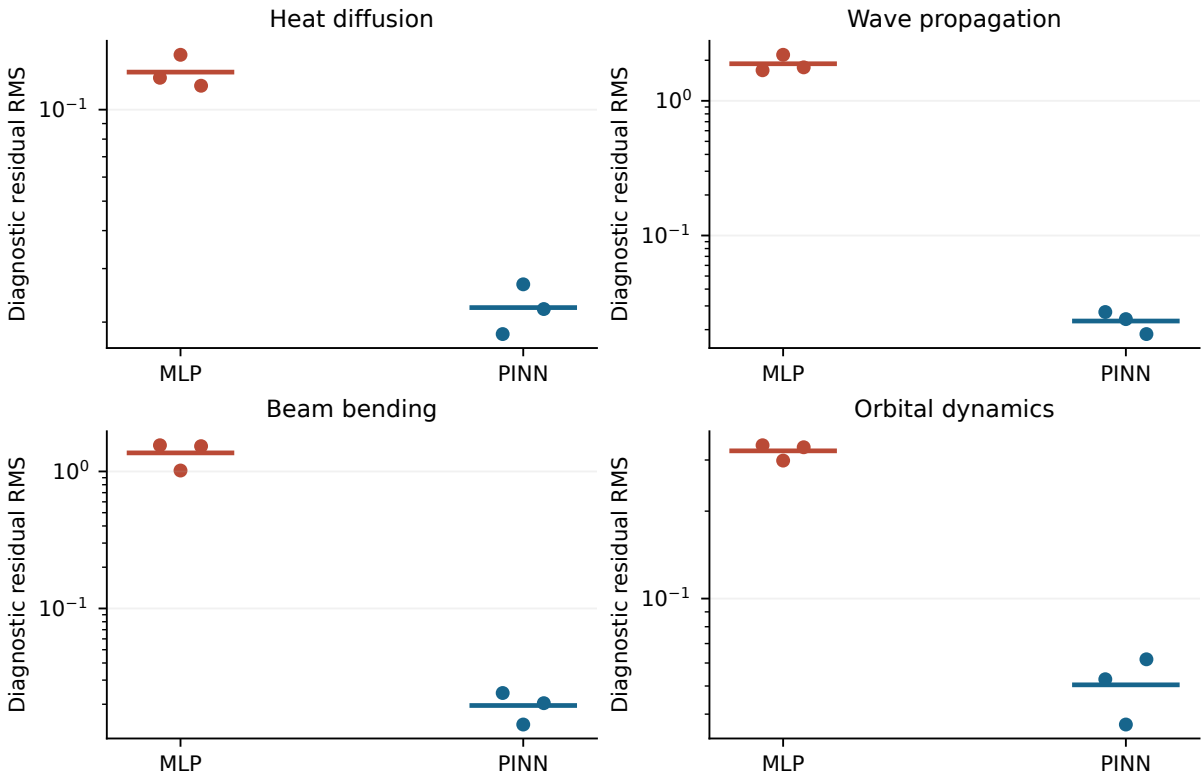


Figure 6: Equation residual RMS at 15% observations, evaluated on 512 coordinates drawn with a separate fixed random seed. The diagnostic sample is not strictly disjoint from training collocation coordinates. Each dot is one training seed, and horizontal bars show means. For the orbit, the value shown is the root mean square across its two residual components. Compare models within each domain, not across equations with different operators and scales.

6 Discussion

6.1 What the comparison shows

The paired design measures the effect of adding the residual term under an equal optimization-step budget. The models were not tuned separately, and no numerical-solver baseline was tested. The wave and orbit results at 15% and 30% show that a data-only network can reconstruct these smooth references more accurately under the chosen protocol. Conversely, the heat results show a consistent benefit from the penalty across all tested observation fractions and seeds. These differences require a domain-specific comparison rather than an assumption that adding physics will improve prediction.

The benchmark fractions are relative to candidate grids of different sizes. Five beam observations and 31 heat observations are both labeled 5%, but they do not carry the same amount of information. The results therefore compare the two models within each domain; they do not rank the domains against each other. High scores on these smooth, low-dimensional reference functions do not establish performance for turbulent flow, sharp shocks, changing geometry or real measurements.

6.2 Residuals, identifiability and computational cost

An equation residual and a reconstruction error answer different questions. The residual measures the local differential relationship. The reconstruction error tests whether the network found the particular state the reference describes. When boundary or initial information is missing, several states can satisfy the equation, so a low residual can accompany a high prediction error even when

automatic differentiation is correct.

Computing fourth derivatives for the beam, and second derivatives and nonlinear acceleration terms for the orbit, adds work to each PINN step. The reported times describe that cost under concurrent CPU execution; they do not support a hardware-independent speed claim. Changing the model size, collocation count, loss weight, initial-condition constraints or optimizer could change the ranking and would require a new controlled comparison.

6.3 Relationship to the application

The live application has domain selectors, training histories and interactive views comparing references with predictions. Existing project screenshots and the two standalone orbital HTML pages document that interface. The predictions embedded in them are not used as evidence in this benchmark. Every plot in Figs. 2–6 was generated for this paper; Fig. 7 in the appendix is the only interface capture.

7 Limitations and future work

The study uses four idealized analytical systems, noiseless labels, one setting of each physical parameter and three random seeds. It gives no evidence about real sensor noise, irregular observation availability, coefficient inference, long-horizon extrapolation, calibrated uncertainty or engineering safety. The test grid measures off-grid interpolation within the sampled domain; it is not an out-of-distribution test. Independent spatial locations are also not independent physical trials.

Because the physics weight, model size and step count are fixed, sensitivity to these choices remains open. For the beam, sparse field labels have to identify the homogeneous components, since support moments are not imposed. For the orbit, the guarded residual differs slightly from the unsoftened reference. The full application also includes adaptive dual-head river models and a coupled reservoir system. These were not part of the analytical benchmark, and each needs its own study of solver convergence, physical scaling and held-out prediction.

Several experiments follow from these limitations. Explicit initial and boundary constraints should be added as a controlled ablation. Observation noise and spatially clustered sensors should be tested, the number of seeds increased, and the loss and residual scales varied. The models should also be compared against standard interpolation and numerical-solver baselines. A separate uncertainty experiment should measure empirical coverage before any interval is called calibrated, and gradient statistics should be recorded before any performance difference is attributed to gradient conflict.

8 Conclusion

This paper evaluates a reusable PINN implementation on four differential-equation systems using paired runs. Numerical figures and tables trace back to fixed configurations, disjoint test coordinates and saved numerical artifacts. Under the fixed protocol, the residual penalty lowered mean error for heat at every observation fraction and for every system at 5%. It raised mean error for the wave and the orbit at 15% and 30%, and for the beam at 30%. Implementation reuse, equation satisfaction and accurate reconstruction need separate checks. The benchmark supports reproducible follow-up experiments; it does not establish readiness for field use.

Code and data

The application code is available in the PINN Multi-Physics Explorer repository. The benchmark uses analytical reference solutions and requires no external measurement dataset. Run settings, prediction arrays, final model states and CSV metrics were retained for this study. Appendix A describes these experimental records.

Author contributions

Alhassan Salifu Babamu developed PINN Multi-Physics Explorer.

Acknowledgements

My internship at RAIL inspired this project. After the internship, I contacted Dr. Adam Affum, Senior Researcher at RAIL, obtained approval to use the laboratory and continued the work under his supervision.

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A Experimental records

The saved records identify the implementation, training settings and numerical outputs used in the comparison. Model states are paired with their run configurations; the prediction arrays retain the observations, independent test references and outputs from both models. The software environment and hardware are described in Section 4.4.

Record	Contents
`source_manifest.json`	Source commit and file-level SHA-256 checksums of the working-tree snapshot.
`results/protocol.json`	Fixed hyperparameters, software versions and execution configuration.
`results/analytic_checks.json`	Float64 exact-function checks on the actual residual modules.
`results/*_s*.json`	Per-pair settings, metrics, timings and recorded loss histories.
`results/*predictions.npz`	Observations, selected indices, independent reference values and both predictions.
`results/*.pt`	Final network state dictionaries, with architecture settings recorded in the protocol.
`results/metrics.csv`	One row per output, model, fraction and seed, including RMSE in physical model units.
`figures/01-06`	Vector PDF and 300 dpi PNG versions of the six scientific figures.

Table 3: Source and numerical records retained for the study.

B Complete sparsity and timing summaries

Domain	Fraction	Observations	MLP error	PINN error
Heat	5%	31	0.0324 ± 0.0151	0.0070 ± 0.0048
Heat	15%	93	0.0172 ± 0.0034	0.0067 ± 0.0043
Heat	30%	187	0.0129 ± 0.0016	0.0050 ± 0.0012
Wave	5%	31	0.0479 ± 0.0165	0.0430 ± 0.0142
Wave	15%	93	0.0286 ± 0.0059	0.0351 ± 0.0079
Wave	30%	187	0.0209 ± 0.0028	0.0324 ± 0.0042
Beam	5%	5	0.2805 ± 0.1972	0.0095 ± 0.0008
Beam	15%	15	0.0080 ± 0.0038	0.0051 ± 0.0043
Beam	30%	30	0.0043 ± 0.0022	0.0057 ± 0.0049
Orbit	5%	20	0.0492 ± 0.0479	0.0337 ± 0.0208
Orbit	15%	60	0.0181 ± 0.0025	0.0268 ± 0.0096
Orbit	30%	120	0.0185 ± 0.0023	0.0274 ± 0.0088

Table 4: Test relative L_2 error at every observation fraction: mean $\pm$ sample standard deviation across three seeds. Observation counts refer to locations; an orbital location supplies two position coordinates.

Domain	Parameters	MLP seconds	PINN seconds
Heat	2,241	7.91	18.22
Wave	2,241	5.46	17.80
Beam	2,209	5.10	40.69
Orbit	2,242	7.00	19.27

Table 5: Parameter counts and observed optimization time at 15% observations. Times are three-seed means in seconds under concurrent CPU execution. Both models have identical counts within each system.

C Interface documentation

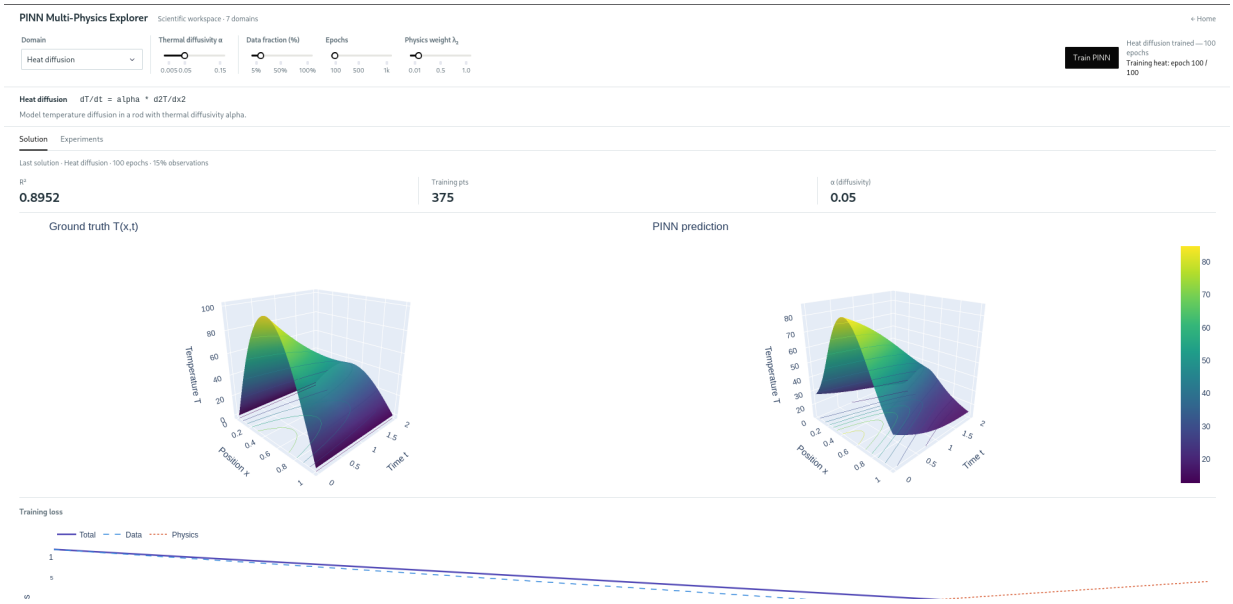


Figure 7: Existing PINN Explorer heat-domain workspace capture, kept from the project portfolio. It shows the application's controls and visualizations. The run displayed is separate from the fixed-protocol experiments in this paper, and its plotted values are not used in any quantitative result.

The repository also contains two visualization pages:

- `examples/orbit_demo.html`
- `threejs_orbit/orbit_scene.html`

These pages are separate from the differential-equation training engine and from the benchmark artifacts in this paper.